

EXAM 7 – SPRING 2012

6. (1.5 points)

Given the following information for an insurance company's book of policies with a \$250,000 deductible:

Age-to-Age Loss and ALAE Development Factors

	<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>	<u>60-72</u>
Limited	1.720	1.151	1.072	1.025	1.011
Excess	2.577	1.412	1.350	1.281	1.316

Limited Severity Relativities (R^L)

<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>	<u>60</u>	<u>72</u>
0.975	0.963	0.955	0.944	0.931	0.912

Calculate the unlimited 36-72 month development factor.

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Question 6 Sample Answer

Solution 1

36-72 dev. factor = 36-48 factor $\times$ 48-60 factor $\times$ 60-72 factor

Limited = $1.072 \times 1.025 \times 1.011 = 1.111$

Excess = $1.350 \times 1.281 \times 1.316 = 2.726$

$$\begin{aligned}\text{LDF}_{36-48} &= \text{LDF}_{36-48}^L \cdot R_{36} + \text{LDF}_{36-48}^{\text{XS}} (1 - R_{36}) \\ &= 1.072(0.955) + 1.350(1-0.955) \\ &= 1.085\end{aligned}$$

$$\begin{aligned}\text{LDF}_{48-60} &= 1.035(0.944) + 1.281(1-0.944) \\ &= 1.039\end{aligned}$$

$$\begin{aligned}\text{LDF}_{60-72} &= 1.011(0.931) + 1.316(1-0.931) \\ &= 1.032\end{aligned}$$

$$\text{total LDF}_{36-72} = 1.088 \times 1.039 \times 1.032 = 1.163$$

Solution 2

Limited $\text{LDF}_{36-72} = (1.072)(1.025)(1.011) = 1.111$

XS $\text{LDF}_{36-72} = (1.35)(1.281)(1.316) = 2.276$

$R^L = 9.55$ – the 36 month rel

$$\begin{aligned}\text{Unlimited LDF}_{36-72} &= (\text{LimLDF}_{36-72}(R^L) + (\text{XSLDF}_{36-72})(1-R^L) \\ &= (1.111) (.955) + (2.276) (1-.955) \\ &= 1.163\end{aligned}$$

Solution 3

From $\frac{\text{LDF}_{36-72}^L}{\text{LDF}_{36-72}} = \frac{R_{72}^L}{R_{36}^L}$, and $\text{LDF}_{36-72}^L = 1.072 \times 1.025 \times 1.011 = 1.1109$

$$R_{72}^L = 0.912 \quad R_{36}^L = 0.955$$

$$\therefore \text{LDF}_{36-72} = \frac{1.1109}{0.912} \times 0.955 = 1.163$$

Solution 4

$$LDF_t = LDF_t^L \times R^L + XSLDF_t^L(1 - R^L)$$

$$LDF_t^L = LDF_t \Delta R^L$$

$$XSLDF_t^L = LDF_t \Delta(1 - R^L)$$

36-72

$$\text{Limited: } 1.072 \times 1.025 \times 1.011 = 1.1108868$$

$$\text{Excess: } 1.35 \times 1.281 \times 1.316 = 2.2758246$$

$$\text{Using } LDF_t^L = LDF_t + \Delta R^L$$

$$\Rightarrow 1.1108868 = LDF_t + (.912/.955)$$

$$\Rightarrow LDF_t = (1.1108/.95497) = \underline{1.163}$$

$$\text{If using } XSLDF_t^L = LDF_t \Delta(1 - R^L)$$

$$2.2758 = LDF_t + ((1 - .912) / (1 - .955)) = LDF_t \times 1.955$$

$$LDF_t = \underline{1.163}$$

Examiner Comment

For this question, most candidates received full credit.

The following was a common error that yielded a correct final factor but used incorrect methodology. We gave partial credits for this response.

$$LDF_t^L = 1.072 \times 1.025 \times 1.011 = 1.1109$$

$$XSLDF_t^L = 1.316 \times 1.281 \times 1.35 = 2.2758$$

$$LDF_t = 1.1109 \times (.912/.955) + 2.2758 \times (1 - .912/.955)$$

$$LDF_{t72} = 1.1633$$